Coffee Sales Analytics

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ABSTRACT

SQL-Based Coffee Sales Analysis Project

- **Project Objective:** To extract meaningful business insights from coffee sales data across Indian cities using SQL queries on a normalized relational database.
- **Analytical Techniques:** Applied SQL operations including JOINs, aggregation, filtering, and grouping to analyze customer behavior, product demand, and city-level performance.
- **Key Findings:** Identified Pune, Chennai, and Bangalore as high-potential markets. Cold Brew and Ground Espresso were top-selling products. Rent-to-revenue ratios guided ROI insights.
- **Strategic Value:** The insights serve as a foundation for data-driven decisions in retail expansion, targeted marketing, inventory planning, and customer segmentation.

COFFEE SALES ANALYTICS PROJECT

SQL-Based Business Insights

- **Project Overview:** This project analyzes coffee product sales data using SQL queries to extract city-wise trends, customer behavior, and revenue patterns across time periods.
- **Database Schema:** Tables include city demographics, customer info, product catalog, and sales records — all linked via foreign keys to support advanced SQL analysis.
- **Key Objectives:** Understand market size, top-selling products, customer segmentation, revenue performance, and affordability trends across Indian cities.
- **Business Relevance:** Insights are designed to help optimize marketing strategy, product stocking, city expansion planning, and customer targeting in the coffee retail industry.

TABLE STRUCTURE AND CONTENT OF THE *CITY* TABLE

Field	Type	Null	Key	Default	Extra
city_id	int	NO	PRI	NULL	
city_name	varchar(15)	YES		NULL	
population	bigint	YES		NULL	
estimated_rent	float	YES		NULL	
city_rank	int	YES		NULL	

city_id	city_name	population	estimated_rent	city_rank
1	Bangalore	12300000	29700	1
2	Chennai	11100000	17100	6
3	Pune	7500000	15300	9
4	Jaipur	4000000	10800	8
5	Delhi	31000000	22500	3
6	Mumbai	20400000	31500	2
7	Hyderabad	10000000	22500	4
8	Ahmedabad	8300000	14400	5
9	Kolkata	14900000	16200	7
10	Surat	7200000	13500	10
11	Lucknow	3800000	9000	11
12	Kanpur	3100000	8100	12
13	Nagpur	2900000	7200	13

TABLE STRUCTURE AND CONTENT OF **PRODUCTS** TABLE

Result Grid | Filter Rows: | Export: | Wrap C

Field	Type	Null	Key	Default	Extra
product_id	int	NO	PRI	NULL	
product_name	varchar(35)	YES		NULL	
Price	float	YES		NULL	

Result Grid | Filter Rows: | Edit:

product_id	product_name	Price
1	Ground Espresso Coffee (250g)	350
2	Cold Brew Coffee Pack (6 Bottles)	900
3	Instant Coffee Powder (100g)	250
4	Coffee Beans (500g)	600
5	Coffee Drip Bags (10 Bags)	450
6	French Press Coffee Set	1200
7	Specialty Coffee Subscription	1500
8	Flavored Coffee Pods (Pack of 10)	750
9	Organic Green Coffee Beans (500g)	700
10	Coffee Gift Hamper	1800
11	Cold Brew Concentrate (500ml)	550
12	Caramel Syrup (250ml)	300
13	Mocha Flavored Coffee Mix (200g)	450

TABLE STRUCTURE AND CONTENT OF THE CUSTOMERS TABLE

Result Grid | Filter Rows: | Export: | Wrap Ce

Field	Type	Null	Key	Default	Extra
customer_id	int	NO	PRI	NULL	
customer_name	varchar(25)	YES		NULL	
city_id	int	YES	MUL	NULL	

Result Grid | Filter Rows:

customer_id	customer_name	city_id
1	Aarav Agarwal	1
2	Aarav Pandey	1
3	Aditi Gupta	1
4	Aditi Joshi	1
5	Aditi Reddy	1
6	Aditi Verma	1
7	Aditya Gupta	1
8	Aditya Malhotra	1
9	Aditya Sharma	1
10	Aditya Singh	1
11	Ananya Gupta	1
12	Ananya Kumar	1
13	Ananya Malhotra	1

TABLE STRUCTURE AND CONTENT OF THE **SALES** TABLE

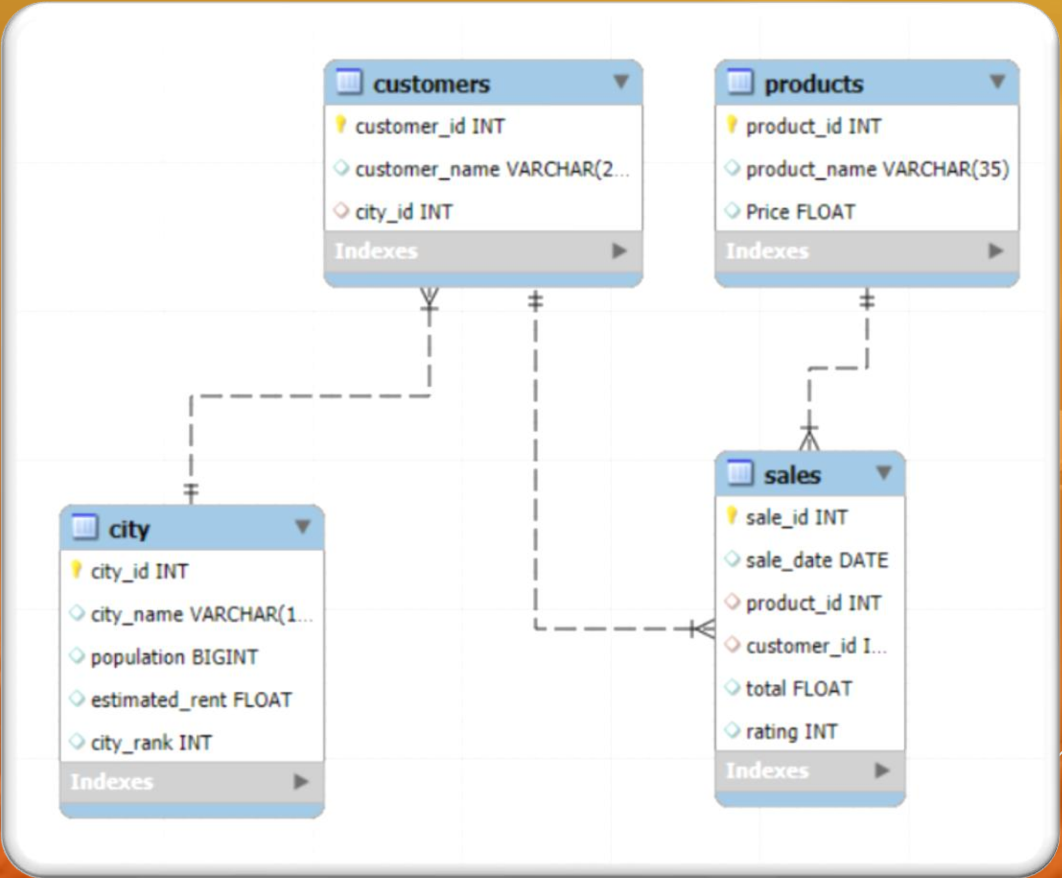
Result Grid | Filter Rows: | Export:

Field	Type	Null	Key	Default	Extra
sale_id	int	NO	PRI	NULL	
sale_date	date	YES		NULL	
product_id	int	YES	MUL	NULL	
customer_id	int	YES	MUL	NULL	
total	float	YES		NULL	
rating	int	YES		NULL	

Result Grid | Filter Rows: | Edit:

sale_id	sale_date	product_id	customer_id	total	rating
1	2023-01-01	2	114	900	4
2	2023-01-01	3	391	250	2
3	2023-01-01	8	168	750	3
4	2023-01-01	15	44	350	4
5	2023-01-01	17	101	600	4
6	2023-01-01	21	330	650	2
7	2023-01-01	28	462	300	3
8	2023-01-02	2	68	900	5
9	2023-01-02	3	109	250	4
10	2023-01-02	4	341	600	2
11	2023-01-02	14	305	320	3
12	2023-01-02	21	110	650	4
13	2023-01-02	22	57	400	5

ER DIAGRAM


customers
customer_id INT
customer_name VARCHAR(2...
city_id INT
Indexes
products
product_id INT
product_name VARCHAR(35)
Price FLOAT
Indexes
city
city_id INT
city_name VARCHAR(1...
population BIGINT
estimated_rent FLOAT
city_rank INT
Indexes
sales
sale_id INT
sale_date DATE
product_id INT
customer_id I...
total FLOAT
rating INT
Indexes


Analysis

"THE ESTIMATED COFFEE CONSUMERS IN EACH CITY ARE CALCULATED BY APPLYING THE ASSUMPTION THAT 25% OF THE TOTAL POPULATION DRINKS COFFEE.

```sql
SELECT
    city_name,
    ROUND(
    (population * 0.25)/1000000,
    2) as coffee_consumers_in_millions,
    city_rank
FROM city
ORDER BY 2 DESC;
```

Result Grid | Filter Rows: | Export

city_name	coffee_consumers_in_millions	city_rank
Delhi	7.75	3
Mumbai	5.10	2
Kolkata	3.73	7
Bangalore	3.08	1
Chennai	2.78	6
Hyderabad	2.50	4
Ahmedabad	2.08	5
Pune	1.88	9
Surat	1.80	10
Jaipur	1.00	8
Lucknow	0.95	11
Indore	0.83	14
Kanpur	0.78	12

ESTIMATED COFFEE CONSUMERS IN EACH CITY

THIS ANALYSIS ESTIMATES COFFEE DRINKERS USING THE FORMULA: 25% OF CITY POPULATION. DELHI, MUMBAI, AND KOLKATA TOP THE LIST WITH OVER 5 MILLION POTENTIAL CUSTOMERS, PROVIDING STRONG INDICATORS FOR DEMAND FORECASTING AND PRIORITIZING MARKET ENTRY OR EXPANSION.

TOTAL REVENUE FROM COFFEE SALES

Result Grid | Filter Rows:

total_revenue
1963300

```sql
SELECT
    SUM(total) as total_revenue
FROM sales
WHERE
    EXTRACT(YEAR FROM sale_date)  = 2023
    AND
    EXTRACT(quarter FROM sale_date) = 4;
```

ANALYSIS USING JOINS

Understanding the Role of SQL Joins in Analytics

Purpose of Joins in SQL
SQL Joins allow you to combine data from multiple tables based on shared columns, essential for working with normalized databases where information is split for efficiency.

Types of Joins
Common types include INNER, LEFT, RIGHT, FULL, and CROSS JOINs—each offering different ways to match or merge rows depending on data relationships.

Why Joins Matter in Analytics
Joins enable comprehensive queries that retrieve detailed, connected insights—for example, combining customer and sales tables to analyze buying patterns.

Example Use Case
In this project, joins were used to merge city, customer, and sales data—allowing accurate segmentation, revenue analysis, and product preference insights.

THE TOTAL REVENUE GENERATED FROM COFFEE SALES ACROSS ALL CITIES IN THE LAST QUARTER OF 2023

```sql
SELECT
    ci.city_name,
    SUM(s.total) as total_revenue
FROM sales as s
JOIN customers as c
ON s.customer_id = c.customer_id
JOIN city as ci
ON ci.city_id = c.city_id
WHERE
    EXTRACT(YEAR FROM s.sale_date)  = 2023
    AND
    EXTRACT(quarter FROM s.sale_date) = 4
GROUP BY 1
ORDER BY 2 DESC;
```

Result Grid | Filter Ro

city_name	total_revenue
Pune	434330
Chennai	302500
Bangalore	270780
Jaipur	248580
Delhi	238490
Kanpur	71890
Mumbai	71340
Surat	52560
Kolkata	51180
Nagpur	45810
Indore	45670
Hyderabad	45060
Ahmedabad	43560
Lucknow	41550

TOTAL REVENUE FROM COFFEE SALES

UNE GENERATED THE HIGHEST REVENUE IN Q4 2023 (₹434,330), FOLLOWED BY CHENNAI AND BANGALORE. THESE RESULTS HIGHLIGHT PUNE AS A HIGHLY LUCRATIVE MARKET DESPITE A SMALLER CONSUMER BASE COMPARED TO DELHI, SIGNALING STRONG PER-CUSTOMER SALES.

SALES COUNT FOR EACH PRODUCT

```sql
SELECT
    p.product_name,
    COUNT(s.sale_id) as total_orders
FROM products as p
LEFT JOIN
sales as s
ON s.product_id = p.product_id
GROUP BY 1
ORDER BY 2 DESC;
```

product_name	total_orders
Cold Brew Coffee Pack (6 Bottles)	1326
Ground Espresso Coffee (250g)	1271
Instant Coffee Powder (100g)	1226
Coffee Beans (500g)	1218
Tote Bag with Coffee Design	776
Vanilla Coffee Syrup (250ml)	762
Cold Brew Concentrate (500ml)	312
Organic Green Coffee Beans (500g)	307
Coffee Art Print	296
Flavored Coffee Pods (Pack of 10)	295
Coffee Drip Bags (10 Bags)	289
Insulated Travel Mug	273
Coffee Gift Hamper	270

SALES COUNT FOR EACH PRODUCT

THE COLD BREW COFFEE PACK (6 BOTTLES) RECORDED THE MOST ORDERS (1,326), FOLLOWED CLOSELY BY GROUND ESPRESSO COFFEE (1,271) AND INSTANT COFFEE POWDER (1,226). THESE TOP SELLERS SHOULD GUIDE INVENTORY DECISIONS AND MARKETING FOCUS.

AVERAGE SALES AMOUNT PER CITY

```sql
SELECT
    ci.city_name,
    SUM(s.total) AS total_revenue,
    COUNT(DISTINCT s.customer_id) AS total_cx,
    ROUND(
        SUM(s.total) / COUNT(DISTINCT s.customer_id)
    , 2) AS avg_sale_pr_cx
FROM sales s
JOIN customers c
    ON s.customer_id = c.customer_id
JOIN city ci
    ON ci.city_id = c.city_id
GROUP BY ci.city_name
ORDER BY total_revenue DESC;
```

Result Grid | Filter Rows: | Export

city_name	total_revenue	total_cx	avg_sale_pr_cx
Pune	1258290	52	24197.88
Chennai	944120	42	22479.05
Bangalore	860110	39	22054.1
Jaipur	803450	69	11644.2
Delhi	750420	68	11035.59
Mumbai	235000	27	8703.7
Kanpur	213550	35	6101.43
Surat	176540	27	6538.52
Kolkata	171460	28	6123.57
Nagpur	140050	24	5835.42
Indore	138590	21	6599.52
Ahmedabad	137690	23	5986.52
Hyderabad	131520	21	6262.86
Lucknow	109400	21	5209.52

AVERAGE SALES AMOUNT PER CITY

PUNE, CHENNAI, AND BANGALORE HAVE THE HIGHEST AVERAGE SALES PER CUSTOMER — OVER ₹22,000 — REVEALING STRONG INDIVIDUAL SPENDING BEHAVIOR. THIS MAKES THESE CITIES PRIME CANDIDATES FOR PREMIUM PRODUCT PROMOTIONS.

CITY POPULATION AND COFFEE CONSUMERS (25%)

```sql
WITH city_table as
(
    SELECT
        city_name,
        ROUND((population * 0.25)/1000000, 2) as coffee_consumers
    FROM city
),
customers_table
AS
(
    SELECT
        ci.city_name,
        COUNT(DISTINCT c.customer_id) as unique_cx
    FROM sales as s
    JOIN customers as c
    ON c.customer_id = s.customer_id
    JOIN city as ci
    ON ci.city_id = c.city_id
     GROUP BY 1
)
SELECT
    customers_table.city_name,
    city_table.coffee_consumers as coffee_consumer_in_millions,
    customers_table.unique_cx
FROM city_table
JOIN
customers_table
ON city_table.city_name = customers_table.city_name;
```

Result Grid | Filter Rows: | Export:

city_name	coffee_consumer_in_millions	unique_cx
Bangalore	3.08	39
Chennai	2.78	42
Pune	1.88	52
Jaipur	1.00	69
Delhi	7.75	68
Mumbai	5.10	27
Hyderabad	2.50	21
Ahmedabad	2.08	23
Kolkata	3.73	28
Surat	1.80	27
Lucknow	0.95	21
Kanpur	0.78	35
Nagpur	0.73	24
Indore	0.83	21

CITY POPULATION AND COFFEE CONSUMERS (25%)

BANGALORE AND CHENNAI NOT ONLY HAVE HIGH ESTIMATED CONSUMER COUNTS BUT ALSO STRONG CUSTOMER ENGAGEMENT. THIS SLIDE CONFIRMS THE ALIGNMENT BETWEEN POTENTIAL MARKET SIZE AND ACTUAL UNIQUE CUSTOMER INTERACTION.

TOP SELLING PRODUCTS BY CITY

```sql
SELECT *
FROM (
    SELECT
        ci.city_name,
        p.product_name,
        COUNT(s.sale_id) AS total_orders,

        DENSE_RANK() OVER(
            PARTITION BY ci.city_name
            ORDER BY COUNT(s.sale_id) DESC
        ) AS rnk

    FROM sales s
    JOIN products p
        ON s.product_id = p.product_id
    JOIN customers c
        ON c.customer_id = s.customer_id
    JOIN city ci
        ON ci.city_id = c.city_id
    GROUP BY ci.city_name, p.product_name
) AS t1
WHERE rnk <= 3
ORDER BY city_name, rnk;
```

Result Grid | Filter Rows: | Export: | Wrap

city_name	product_name	total_orders	rnk
Ahmedabad	Cold Brew Coffee Pack (6 Bottles)	40	1
Ahmedabad	Coffee Beans (500g)	35	2
Ahmedabad	Instant Coffee Powder (100g)	26	3
Bangalore	Cold Brew Coffee Pack (6 Bottles)	197	1
Bangalore	Ground Espresso Coffee (250g)	167	2
Bangalore	Instant Coffee Powder (100g)	150	3
Chennai	Cold Brew Coffee Pack (6 Bottles)	192	1
Chennai	Coffee Beans (500g)	181	2
Chennai	Instant Coffee Powder (100g)	172	3
Delhi	Ground Espresso Coffee (250g)	183	1
Delhi	Instant Coffee Powder (100g)	170	2
Delhi	Coffee Beans (500g)	161	3
Hyderabad	Instant Coffee Powder (100g)	36	1
Hyderabad	Cold Brew Coffee Pack (6 Bottles)	28	2
Hyderabad	Ground Espresso Coffee (250g)	27	3

TOP SELLING PRODUCTS BY CITY

PRODUCT PREFERENCES VARY BY REGION. FOR EXAMPLE, BANGALORE LEADS WITH COLD BREW (197 ORDERS), WHILE AHMEDABAD'S TOP PRODUCT IS ALSO COLD BREW (40 ORDERS). THESE INSIGHTS SUPPORT CITY-SPECIFIC STOCK AND AD STRATEGIES

CUSTOMER SEGMENTATION BY CITY

```sql
SELECT
    ci.city_name,
    COUNT(DISTINCT c.customer_id) as unique_cx
FROM city as ci
LEFT JOIN
customers as c
ON c.city_id = ci.city_id
JOIN sales as s
ON s.customer_id = c.customer_id
WHERE
    s.product_id IN (1, 2, 3, 4, 5, 6, 7, 8, 9,
    10, 11, 12, 13, 14)
GROUP BY 1;
```

Result Grid | Filter Ro

city_name	unique_cx
Ahmedabad	23
Bangalore	39
Chennai	42
Delhi	68
Hyderabad	21
Indore	21
Jaipur	69
Kanpur	35
Kolkata	28
Lucknow	21
Mumbai	27
Nagpur	24
Pune	52
Surat	27

CUSTOMER SEGMENTATION BY CITY

CHENNAI (42), BANGALORE (39), AND DELHI (68) HAVE THE HIGHEST UNIQUE CUSTOMER COUNTS. THESE NUMBERS REFLECT STRONG CUSTOMER REACH, WITH SEGMENTATION DATA SUPPORTING HYPER-TARGETED CAMPAIGNS AND LOYALTY PROGRAMS.

AVERAGE SALE VS RENT

```sql
WITH city_table AS (
    SELECT
        ci.city_name,
        SUM(s.total) AS total_revenue,
        COUNT(DISTINCT s.customer_id) AS total_cx,
        ROUND(
            SUM(s.total) * 1.0 / COUNT(DISTINCT s.customer_id)
        , 2) AS avg_sale_pr_cx
    FROM sales AS s
    JOIN customers AS c
        ON s.customer_id = c.customer_id
    JOIN city AS ci
        ON ci.city_id = c.city_id
    GROUP BY ci.city_name
),
city_rent AS (
    SELECT
        city_name,
        estimated_rent
    FROM city
)
SELECT
    cr.city_name,
    cr.estimated_rent,
    ct.total_cx,
    ct.avg_sale_pr_cx,
    ROUND(
        cr.estimated_rent * 1.0 / ct.total_cx
    , 2) AS avg_rent_per_cx
FROM city_rent AS cr
JOIN city_table AS ct
    ON cr.city_name = ct.city_name
ORDER BY ct.avg_sale_pr_cx DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

city_name	estimated_rent	total_cx	avg_sale_pr_cx	avg_rent_per_cx
Pune	15300	52	24197.88	294.23
Chennai	17100	42	22479.05	407.14
Bangalore	29700	39	22054.1	761.54
Jaipur	10800	69	11644.2	156.52
Delhi	22500	68	11035.59	330.88
Mumbai	31500	27	8703.7	1166.67
Indore	6300	21	6599.52	300
Surat	13500	27	6538.52	500
Hyderabad	22500	21	6262.86	1071.43
Kolkata	16200	28	6123.57	578.57
Kanpur	8100	35	6101.43	231.43
Ahmedabad	14400	23	5986.52	626.09
Nagpur	7200	24	5835.42	300
Lucknow	9000	21	5209.52	428.57

AVERAGE SALE VS RENT

PUNE SHOWS THE BEST PERFORMANCE WITH THE HIGHEST AVERAGE SALE PER CUSTOMER (₹24,197) AND RELATIVELY LOW RENT (₹294 PER CUSTOMER). CHENNAI AND BANGALORE FOLLOW. THIS ANALYSIS HELPS PRIORITIZE HIGH-ROI CITIES FOR EXPANSION.

MONTHLY SALES GROWTH

```sql
WITH monthly_sales AS (
    SELECT
        ci.city_name,
        MONTH(s.sale_date) AS month,
        YEAR(s.sale_date)  AS year,
        SUM(s.total)       AS total_sale
    FROM sales AS s
    JOIN customers AS c
        ON c.customer_id = s.customer_id
    JOIN city AS ci
        ON ci.city_id = c.city_id
    GROUP BY
        ci.city_name,
        YEAR(s.sale_date),
        MONTH(s.sale_date)
),
growth_ratio AS (
    SELECT
        city_name,
        month,
        year,
        total_sale AS cr_month_sale,
        LAG(total_sale, 1) OVER (
            PARTITION BY city_name
            ORDER BY year, month
        ) AS last_month_sale
    FROM monthly_sales
)
```

```sql
SELECT
    city_name,
    month,
    year,
    cr_month_sale,
    last_month_sale,
    ROUND(
        (cr_month_sale - last_month_sale) * 100.0
        / NULLIF(last_month_sale, 0)
    , 2) AS growth_ratio
FROM growth_ratio
WHERE last_month_sale IS NOT NULL
ORDER BY city_name, year, month;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

city_name	month	year	cr_month_sale	last_month_sale	growth_ratio
Ahmedabad	2	2024	10900	12090	-9.84
Ahmedabad	3	2024	14000	10900	28.44
Ahmedabad	4	2024	3950	14000	-71.79
Ahmedabad	5	2024	5250	3950	32.91
Ahmedabad	6	2024	3300	5250	-37.14
Ahmedabad	7	2024	2700	3300	-18.18
Ahmedabad	8	2024	3550	2700	31.48
Ahmedabad	9	2024	2650	3550	-25.35
Bangalore	2	2023	24750	36890	-32.91
Bangalore	3	2023	26120	24750	5.54
Bangalore	4	2023	23520	26120	-9.95
Bangalore	5	2023	37790	23520	60.67
Bangalore	6	2023	37790	37790	0
Bangalore	7	2023	33120	37790	-12.36
Bangalore	8	2023	31050	33120	-6.25

MONTHLY SALES GROWTH

HMEDABAD SHOWS INCONSISTENT BUT OCCASIONALLY STRONG GROWTH (E.G., +32.91% IN MAY), WHILE BANGALORE EXPERIENCED SHARP DROPS (E.G., -32.91% IN MARCH 2023). THESE TRENDS SUGGEST THE NEED FOR DEMAND STABILIZATION STRATEGIES OR TIMING PROMOTIONS.

BUSINESS METRICS FOR TOP 3 PERFORMING CITIES

```sql
WITH city_table AS (
    SELECT
        ci.city_name,
        SUM(s.total) AS total_revenue,
        COUNT(DISTINCT s.customer_id) AS total_cx,
        ROUND(
            SUM(s.total) * 1.0 / COUNT(DISTINCT s.customer_id)
        , 2) AS avg_sale_pr_cx
    FROM sales AS s
    JOIN customers AS c
        ON s.customer_id = c.customer_id
    JOIN city AS ci
        ON ci.city_id = c.city_id
    GROUP BY ci.city_name
),
city_rent AS (
    SELECT
        city_name,
        estimated_rent,
        ROUND(population * 0.25 / 1000000, 3) AS estimated_coffee_consumer_in_millions
    FROM city
)
SELECT
    cr.city_name,
    ct.total_revenue,
    cr.estimated_rent AS total_rent,
    ct.total_cx,
    cr.estimated_coffee_consumer_in_millions,
    ct.avg_sale_pr_cx,
    ROUND(
        cr.estimated_rent * 1.0 / ct.total_cx
    , 2) AS avg_rent_per_cx
FROM city_rent AS cr
JOIN city_table AS ct
    ON cr.city_name = ct.city_name
ORDER BY ct.total_revenue DESC
LIMIT 3;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

city_name	total_revenue	total_rent	total_cx	estimated_coffee_consumer_in_millions	avg_sale_pr_cx	avg_rent_per_cx
Pune	1258290	15300	52	1.875	24197.88	294.23
Chennai	944120	17100	42	2.775	22479.05	407.14
Bangalore	860110	29700	39	3.075	22054.1	761.54

BUSINESS METRICS FOR TOP 3 PERFORMING CITIES

PUNE LEADS IN REVENUE, COST EFFICIENCY, AND CUSTOMER BASE. CHENNAI AND BANGALORE ARE STRONG OVERALL, BUT BANGALORE'S HIGHER RENT PER CUSTOMER (₹761.54) LOWERS ROI. THIS COMBINED VIEW GUIDES BALANCED BUSINESS DECISIONS

CONCLUSION & STRATEGIC RECOMMENDATIONS

Summary of Data-Driven Coffee Sales Insights

Pune Leads in Profitability
With the highest average sales per customer and efficient rent, Pune is the most cost-effective city for business growth.

Top Products Identified
Cold Brew, Espresso, and Instant Coffee Powder are the best-sellers and should be prioritized in promotions and stock planning.

High-Value Customer Hubs
Chennai, Bangalore, and Delhi have the highest number of unique customers, ideal for loyalty and personalization strategies.

Growth and Inventory Timing
Cities like Ahmedabad show seasonal demand swings; timing promotions and inventory based on monthly trends is key.

THANK YOU